MODULE 1 · LECTURE 2

# Anatomy of Computers: Hardware, Software and Input/Output Devices

*The parts inside a computer, how the processor works, the kinds of software, and the devices that bring data in and send results out — with farm examples.*

### Learning objectives

After this lecture you should be able to:

1. Describe the main hardware parts of a computer and what each does.
2. Explain the parts of the CPU and the fetch–decode–execute–store cycle.
3. Explain buses, ports and how address-bus width limits memory.
4. Classify software into system and application software, and compare compilers, interpreters and assemblers.
5. Describe common input and output devices, including those used on farms and in agri-business.
6. Choose a suitable computer or field device for an agricultural task.

Scope: Lecture 1 introduced the block diagram of a computer. This lecture opens each block. Memory and storage devices are covered in Lecture 3, operating systems in Lecture 4 and programming languages in Lecture 8.

## 1. The computer system

A working computer system has four parts: **hardware** (the physical machine), **software** (the instructions), **data** (what is processed) and **people** (users, operators and programmers). A useful comparison is the human body: hardware is the body, software is the knowledge that tells it what to do, and data is what it works on. Hardware without software cannot do anything; software without hardware has nothing to run on.

## 2. Hardware inside the system unit

| Part | What it does |
|---|---|
| Motherboard | The main circuit board. It holds the CPU socket, memory (RAM) slots, the chipset that manages data flow, expansion slots, the firmware chip and the ports, and connects every part to every other |
| Central processing unit (CPU) | Executes the instructions of programs — the "brain" of the computer (Section 3) |
| Memory (RAM) modules | Hold the programs and data in use (Lecture 3) |
| Storage drives | Hard disk drive (HDD) or solid-state drive (SSD) for permanent storage (Lecture 3) |
| Power supply unit (SMPS) | Converts mains alternating current into the low direct-current voltages the electronics need |
| Cooling system | Fans and heat sinks remove the heat produced by the processor and power supply |
| Expansion cards | Add functions — for example a graphics card (GPU) or a network card |
| Firmware chip (BIOS/UEFI) | Stores the start-up program that tests the hardware and loads the operating system |

## 3. The central processing unit

The CPU has three main parts (Patterson & Hennessy, 2014; Rajaraman & Adabala, 2015):

- **Control unit (CU)** — fetches each instruction from memory, decodes it and sends control signals to the other units so they act in the right order.
- **Arithmetic–logic unit (ALU)** — performs arithmetic (add, subtract, multiply, divide) and logical operations (compare, AND, OR, NOT).
- **Registers** — a few very fast storage locations inside the CPU. Important ones are the **program counter (PC)**, which holds the address of the next instruction; the **instruction register (IR)**, which holds the instruction being executed; the **accumulator**, which holds results of the ALU; and the **memory address register (MAR)** and **memory data register (MDR)**, which hold the address and the data being moved to or from memory.

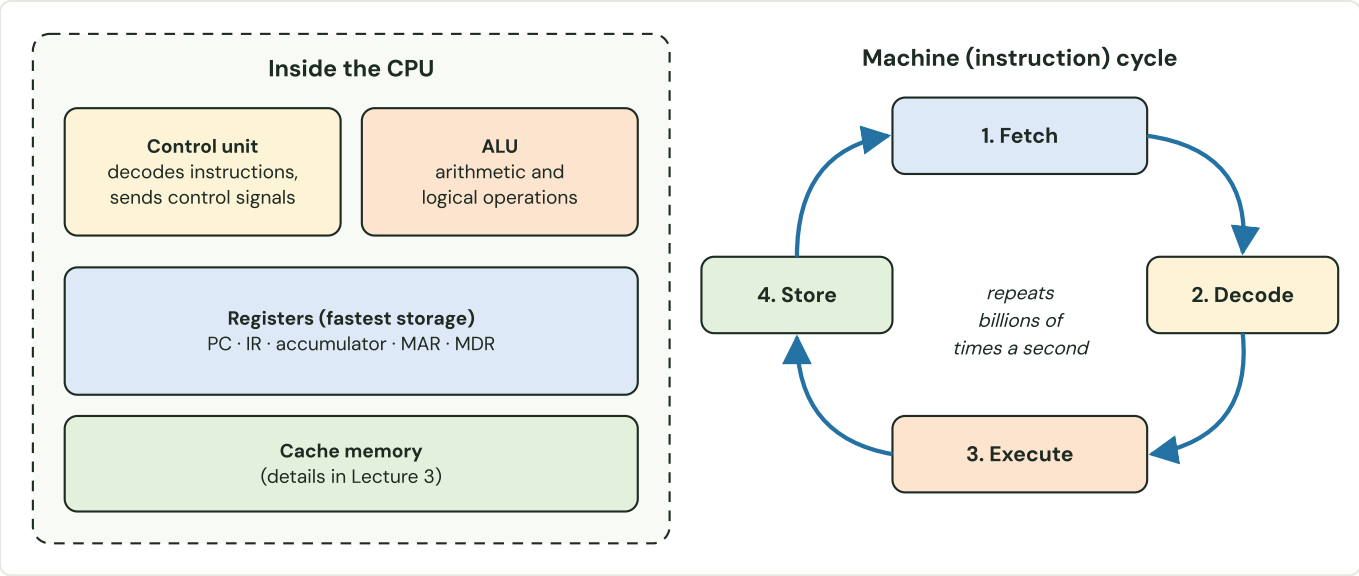



*Figure 1. Parts of the CPU and the machine cycle. Original diagram.*

## The machine cycle

1. **Fetch** — the control unit reads the next instruction from the memory address in the program counter.
2. **Decode** — it works out what the instruction means and which data it needs.
3. **Execute** — the ALU (or another unit) carries out the operation.
4. **Store** — the result is written to a register or to memory, and the program counter moves on.

## Processor speed and size

- **Clock speed** is the number of cycles the processor clock completes per second, measured in hertz (Hz). A 3 GHz processor has $3 \times 10^9$ cycles per second.
- **Word length** is the number of bits the CPU handles as one unit — most modern computers are 64-bit.
- **Multi-core processors** put two or more processing cores on one chip so that several tasks run at the same time.
- **Graphics processing units (GPUs)** contain hundreds or thousands of simpler cores that work in parallel. They draw graphics and also speed up artificial-intelligence work such as training crop-disease image models (Lectures 19–28).
- **Microprocessor vs microcontroller** — a microprocessor is the CPU of a general-purpose computer; a microcontroller combines a small CPU, memory and input/output on one chip and is used in embedded devices such as tractor engine control units and irrigation controllers.

**Worked example — what does 3 GHz mean?**
One clock cycle lasts $1/(3 \times 10^9)$ s = $0.33 \times 10^{-9}$ s ≈ 0.33 ns.
If an instruction needs on average 4 cycles, one core completes $3 \times 10^9 / 4 = 750$ million instructions per second (750 MIPS).
A 4-core processor running four such programs at once can, in the best case, complete about $4 \times 750 = 3000$ MIPS.
(Real processors overlap the steps of many instructions, so actual figures vary; the example only shows how the units relate.)

## 4. Buses and ports

A **bus** is a set of parallel electrical paths that carries signals between the CPU, memory and input/output devices. A computer has three kinds:

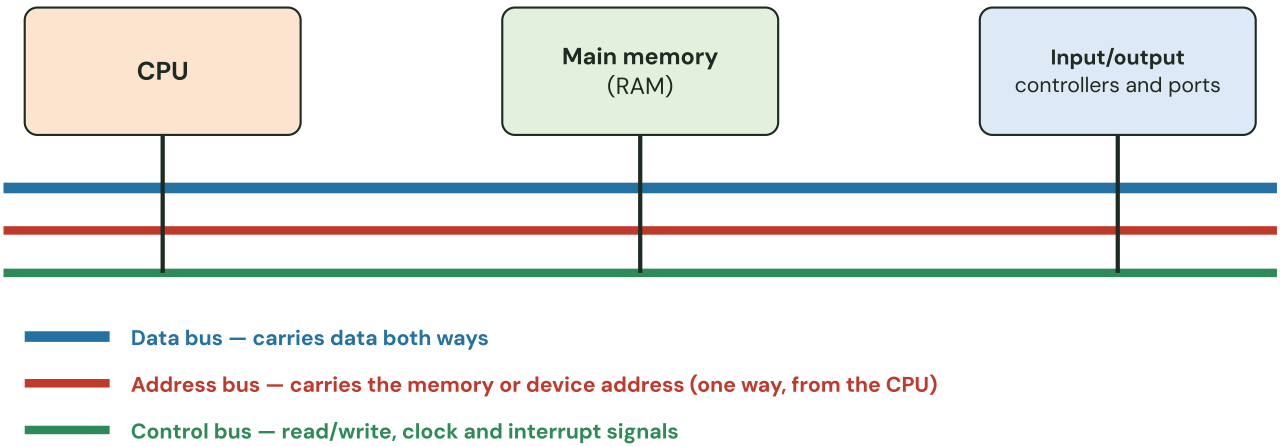



*Figure 2. The three buses that link the CPU, memory and input/output. Original diagram.*

- **Data bus** — carries the data itself, in both directions. Its width (e.g. 64 bits) sets how much data moves at once.
- **Address bus** — carries the address of the memory location or device the CPU wants. With n address lines the CPU can address $2^n$ locations.
- **Control bus** — carries timing and command signals such as read, write, clock and interrupt.

**Worked example — address-bus width and memory**
With 32 address lines the CPU can address $2^{32}$ = 4,294,967,296 locations. If each location holds one byte, that is 4 GiB of memory — the reason 32-bit computers could not use more than about 4 GB of RAM. With 64-bit addressing the limit is far beyond any memory fitted today. (Units of memory are explained in Lecture 3.)

### Ports and connectors

| Port / link | Used for | Farm or field example |
|---|---|---|
| USB (Universal Serial Bus) | Keyboards, mice, printers, storage, phones; also supplies power | Downloading readings from a weather-station data logger |
| HDMI / DisplayPort / VGA | Monitors and projectors (VGA is the older analogue type) | Showing a crop-advisory dashboard in a training hall |
| Ethernet (RJ-45) | Wired network connection | Connecting a dairy-plant control computer to the office network |
| Audio jacks | Speakers, headphones and microphones | Recording voice queries for an advisory service |
| Serial links (RS-232, RS-485) | Industrial sensors and controllers | Soil and weather sensors wired to a controller |
| CAN bus / ISOBUS | Networks inside vehicles and machines; ISOBUS (ISO 11783) lets a tractor terminal control implements from different makers | A tractor terminal controlling a variable-rate seed drill |
| Wi-Fi, Bluetooth, mobile data (4G/5G) | Wireless connections | A soil-moisture probe sending data to a phone app |

## 5. Software

Software is the set of programs, data and instructions that tells the hardware what to do. A **program** is an ordered set of instructions written to perform a task. Software is broadly divided into **system software**, which runs and manages the computer itself, and **application software**, which does useful work for the user.

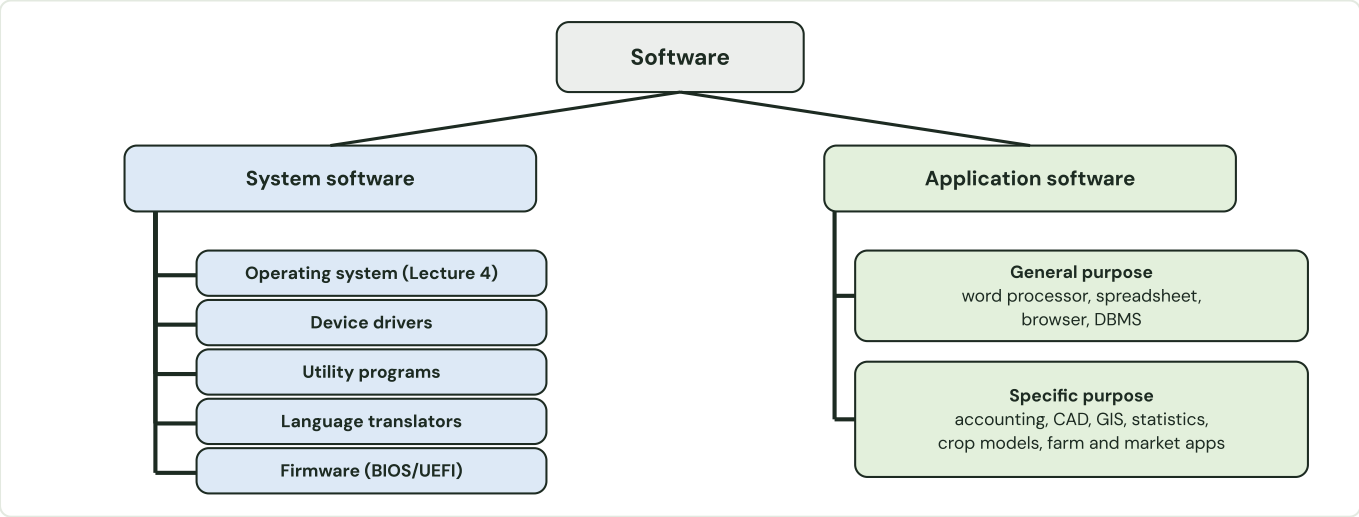



*Figure 3. Classification of software. Original diagram.*

### System software

| Type | Purpose | Examples |
|---|---|---|
| Operating system | Manages hardware, memory, files and programs and provides the user interface (Lecture 4) | Windows, Linux, macOS, Android |
| Device drivers | Let the operating system communicate with a particular device | Printer driver, USB-to-serial driver for a data logger |
| Utility programs | Maintain and protect the computer | Antivirus, file compression, backup, disk clean-up |
| Language translators | Convert programs written by people into machine code | Assembler, compiler, interpreter |
| Firmware | Software stored permanently in a chip on the hardware | BIOS/UEFI of a PC; the program inside a drip controller |

| Translator | Translates | How | Notes |
|---|---|---|---|
| Assembler | Assembly language → machine code | Whole program | One assembly instruction usually becomes one machine instruction |
| Compiler | High-level language (C, C++, Java) → machine code | Whole program at once, before running | Fast to run; lists all errors after compiling |
| Interpreter | High-level language (Python, BASIC) → actions | One statement at a time, while running | Easy to test and debug; usually slower to run |

### Application software

- **General-purpose** — word processors, spreadsheets, presentation software, database programs and web browsers (Lectures 5–7).
- **Specific-purpose** — accounting, computer-aided design (CAD), geographic information systems (GIS), statistical packages, and agricultural tools such as FAO's CROPWAT 8.0, which calculates crop water and irrigation requirements from soil, climate and crop data (FAO, n.d.), and crop simulation models such as DSSAT (Jones et al., 2003).
- **Mobile apps** — weather, market-price and advisory apps on smartphones (Lecture 14).

### Software by licence

| Licence type | Meaning | Example |
|---|---|---|
| Proprietary (commercial) | Bought or licensed; source code is closed | Microsoft Office |
| Freeware | Free to use but source code is closed | Many free mobile apps |
| Shareware / trial | Free for a trial period, then paid | Trial versions of design software |
| Open-source | Source code is available; users may study, change and share it under the licence | Linux, LibreOffice, QGIS, R |

## 6. Input devices

Input devices convert data and commands from the outside world into a form the computer can process.

| Device | How it works | Agricultural or rural use |
|---|---|---|
| Keyboard | Keys for letters, numbers, functions and navigation; the QWERTY layout comes from 19th-century typewriters | Entering farm records and survey data |
| Mouse and touchpad | Moves a pointer on the screen; the mouse was developed at SRI by Douglas Engelbart — Bill English built the first prototype in 1964 and it was shown publicly in 1968 (SRI International, n.d.) | Working with maps and spreadsheets |
| Trackball, joystick, light pen, stylus | Other pointing devices | Stylus on a tablet for field mapping |
| Touch screen | Detects a finger or stylus on the display (resistive or capacitive) | Tractor terminals, kiosks, smartphones |
| Scanner | Converts a paper document or image into a digital image | Digitising land records and old field notebooks |
| OCR (optical character recognition) | Software that reads printed text in a scanned image and turns it into editable text | Converting printed reports into searchable files |
| OMR (optical mark recognition) | Reads pencil or pen marks in fixed boxes | Examination answer sheets; farm-survey forms |
| MICR (magnetic ink character recognition) | Reads characters printed in magnetic ink; Indian cheques carry a MICR band at the bottom printed in the E-13B font (Reserve Bank of India, n.d.) | Processing cheques in rural banks and cooperatives |
| Barcode reader | Reads one-dimensional barcodes (lines of varying width) | Seed packets, fertiliser bags, soil-sample labels |
| QR code reader | Reads two-dimensional QR codes, released by DENSO WAVE in 1994 (DENSO WAVE, n.d.); any smartphone camera can read them | Product traceability, payment at mandis and shops |
| RFID reader | Reads radio-frequency identification tags without line of sight | Animal identification in dairy herds |
| Biometric devices | Read fingerprints, iris or face | Identity checks for scheme benefits |
| Microphone | Captures sound and speech | Voice queries in local languages |
| Digital camera / webcam | Captures images and video | Photographing diseased leaves for diagnosis apps (Lecture 25) |
| Digitiser (graphics tablet) | Traces drawings and maps into coordinates | Digitising field and village maps (Lecture 15) |
| Sensors and GNSS receivers | Convert physical quantities (moisture, temperature, rain, position) into electrical signals | Weather stations, soil-moisture probes, GPS guidance |

**Agricultural example — Pashu Aadhaar**
Under India's livestock-identification programme, each animal receives a **12-digit, bar-coded ear tag** (Pashu Aadhaar). Field workers read or enter the number with a handheld device or smartphone, and the record is linked to the animal's breeding, vaccination and treatment history in the national database (Joshi, 2023). The ear tag is the input medium; the barcode reader or phone is the input device.

## 7. Output devices

Output devices convert processed results into a form people (or machines) can use. Output on a screen is **soft copy**; output on paper is **hard copy**.

### Monitors

- **Types:** cathode-ray tube (CRT, now obsolete), liquid-crystal display (LCD), LED-backlit LCD and organic LED (OLED).
- **Screen size** is measured diagonally (e.g. 24-inch).
- **Resolution** is the number of picture elements (**pixels**) across and down, e.g. 1920 × 1080 ("Full HD").
- **Refresh rate** (Hz) is how many times per second the picture is redrawn.

**Worked example — how much data is one screen image?**
Pixels = 1920 × 1080 = 2,073,600.
With 24-bit colour each pixel needs 3 bytes, so one frame = 2,073,600 × 3 = 6,220,800 bytes ≈ 6.2 MB.
This is why images and video need compression and large storage (Lecture 3).

### Printers and plotters

| Type | How it works | Typical use |
|---|---|---|
| Dot-matrix (impact) | Pins strike an inked ribbon | Multi-part forms and receipts |
| Inkjet (non-impact) | Sprays tiny drops of ink | Colour photos and reports in small offices |
| Laser (non-impact) | A laser draws the page on a light-sensitive drum, which picks up toner that is fused onto paper; Gary Starkweather built the first laser printer at Xerox PARC after moving there in 1971 (Computer History Museum, n.d.) | Fast, sharp office printing |
| Thermal | Heat darkens special paper | Weighbridge slips, mandi receipts, labels |
| Plotter | Draws large drawings with pens or inkjet heads | CAD drawings of implements, large maps |
| 3D printer | Builds an object layer by layer | Prototype parts for farm machinery |

Print quality is measured in **dots per inch (dpi)**: at 600 dpi a square inch holds 600 × 600 = 360,000 dots. Speed is measured in **pages per minute (ppm)**.

### Other output devices

- **Speakers and headphones** — sound and voice output, such as spoken advisories.

- **Projectors** — large-screen display for training and extension meetings.
- **Actuators** — outputs of embedded computers that act on the physical world: relays, motors, solenoid valves and variable-rate applicators (Lecture 12).

Some devices do both input and output: touch screens, multifunction printer-scanners, network adapters, modems and storage drives.

## 8. Putting it together: an automatic drip-irrigation controller

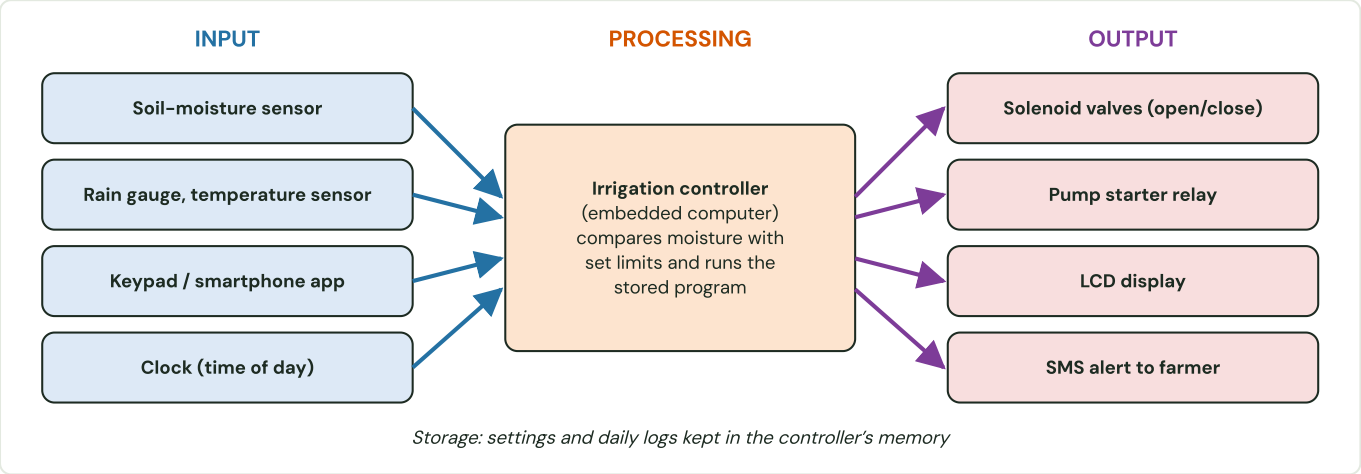



*Figure 4. Input, processing and output in an automatic drip-irrigation system. Original diagram.*

The controller is a small embedded computer. Its sensors and keypad are input devices; its microcontroller runs a stored program that compares soil moisture with the limits the farmer has set; its valves, pump relay, display and SMS messages are outputs. Lecture 13 explains how such systems estimate crop water needs.

## 9. Choosing a computer or field device

| Factor | What to check |
|---|---|
| Processor and RAM | Enough for the software — GIS and image work need more than office work |
| Storage | An SSD makes the computer start and respond faster than an HDD |
| Display | Size, resolution and outdoor readability for field use |
| Ports and wireless | USB, network and Bluetooth for loggers, sensors and printers |
| Battery and power | Battery life for field work; a UPS (uninterruptible power supply) in offices with unreliable mains |
| Ruggedness | Dust and water protection for field devices, given by the IP rating (IEC 60529) — e.g. IP65 is dust-tight and protected against water jets |
| Software | Compatibility with the operating system and the applications needed |
| Cost and support | Price, warranty, local service and spare parts |

**Care and disposal:** protect equipment from dust, heat and voltage surges, and dispose of old computers and batteries through authorised e-waste collectors, as required in India under the E-Waste

(Management) Rules, 2022.

## 10. Key takeaways

- A computer system is hardware, software, data and people working together.
- The motherboard connects the CPU, memory, storage, power supply and expansion cards.
- The CPU's control unit, ALU and registers carry out the fetch–decode–execute–store cycle billions of times a second.
- Data, address and control buses link the CPU, memory and devices; n address lines address $2^n$ locations.
- System software (operating system, drivers, utilities, translators, firmware) runs the computer; application software does the user's work.
- Input devices range from keyboards and scanners to OMR, MICR, barcode, QR, RFID and field sensors.
- Output devices include monitors, printers, plotters, speakers and — in farm automation — actuators such as solenoid valves.

## 11. Examination questions

### Short-answer questions (2 marks)

**Q1. What is a motherboard?**

The main circuit board of a computer, which holds and connects the CPU, memory, chipset, expansion slots, firmware and ports.

**Q2. Name the three main parts of a CPU.**

Control unit, arithmetic–logic unit and registers.

**Q3. What is the function of the program counter?**

It holds the memory address of the next instruction to be fetched.

**Q4. List the steps of the machine cycle.**

Fetch, decode, execute and store.

**Q5. How many memory locations can a CPU with 20 address lines address?**

$2^{20}$ = 1,048,576 locations (1 Mi).

**Q6. Distinguish between a compiler and an interpreter.**

A compiler translates the whole program before it runs; an interpreter translates and executes one statement at a time.

**Q7. What is firmware?**

Software stored permanently in a hardware chip, such as the BIOS/UEFI of a PC or the program in a drip controller.

**Q8. What do OMR and MICR stand for? Give one use of each.**

Optical mark recognition (examination answer sheets) and magnetic ink character recognition (bank cheques).

**Q9. Distinguish between soft copy and hard copy.**

Soft copy is output shown on a screen; hard copy is output printed on paper.

**Q10. What is an actuator? Give a farm example.**

An output device that acts on the physical world under computer control — e.g. a solenoid valve in a drip system.

## Long-answer questions (5–10 marks) — answer outlines

**Q1. Explain the structure of the CPU and the machine cycle with a diagram.**

Draw Figure 1; explain CU, ALU and registers (PC, IR, accumulator, MAR, MDR); describe fetch–decode–execute–store; add clock speed with the 3 GHz example.

**Q2. What are buses? Explain the types of buses.**

Definition; data, address and control buses (Figure 2); bus width; $2^n$ addressing with the 32-bit example.

**Q3. Classify software with examples.**

System software (OS, drivers, utilities, translators, firmware) and application software (general and specific, with agricultural examples such as CROPWAT and DSSAT); licence types.

**Q4. Describe input devices used in computers. Which are useful in agriculture?**

Keyboard, pointing devices, touch screen, scanner, OCR, OMR, MICR, barcode/QR, RFID, biometrics, microphone, camera, digitiser, sensors and GNSS — with examples such as Pashu Aadhaar ear tags and soil-moisture sensors.

**Q5. Describe output devices, with types of printers.**

Monitors (types, size, resolution, refresh rate, the 6.2 MB example), printers (dot-matrix, inkjet, laser, thermal), plotters, 3D printers, speakers, projectors and actuators.

**Q6. Explain input, processing and output in an automatic irrigation controller.**

Use Figure 4: sensors, keypad and clock as inputs; microcontroller and stored program as processing; valves, relay, display and SMS as outputs; memory for logs.

## 12. Glossary

| Term | Meaning |
|---|---|
| Actuator | A device that converts a control signal into physical action (valve, motor, relay) |
| Address bus | Lines that carry the address of a memory location or device |
| BIOS/UEFI | Firmware that starts the computer and loads the operating system |
| Bus | A set of electrical paths carrying signals between parts of a computer |
| Clock speed | Number of processor clock cycles per second, in hertz |
| Compiler | A translator that converts a whole high-level program into machine code before it runs |
| Device driver | Software that lets the operating system control a particular device |
| dpi | Dots per inch — a measure of print or scan resolution |
| Firmware | Software stored permanently in a hardware chip |
| GPU | Graphics processing unit — a processor with many parallel cores |
| Interpreter | A translator that executes a program one statement at a time |
| Machine cycle | Fetch, decode, execute and store — the steps for each instruction |
| Microcontroller | A small computer on one chip, used in embedded devices |
| Motherboard | The main circuit board of a computer |
| OCR | Optical character recognition — reading printed text from images |
| OMR | Optical mark recognition — reading marks in fixed positions |
| MICR | Magnetic ink character recognition — reading magnetic-ink characters on cheques |
| Open-source software | Software whose source code is available to study, change and share |
| Pixel | The smallest picture element on a screen |
| Register | A very fast storage location inside the CPU |
| RFID | Radio-frequency identification — reading tags by radio waves |

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*All diagrams are original. Explanations are written in original language; standard textbook content on CPU parts, buses, software types and devices follows Patterson and Hennessy (2014), Rajaraman and Adabala (2015) and Sinha and Sinha (2004).*

## Practice MCQs

**Q1. The main circuit board of a computer is the**

(a) chipset (b) motherboard
(c) power supply (d) expansion card

**Q2. Which register holds the address of the next instruction?**

(a) Accumulator (b) Instruction register
(c) Program counter (d) Memory data register

**Q3. The correct order of the machine cycle is**

(a) decode, fetch, execute, store (b) fetch, decode, execute, store
(c) fetch, execute, decode, store (d) store, fetch, decode, execute

**Q4. A CPU with 32 address lines can address**

(a) 32 locations (b) $2^{32}$ locations
(c) $32^2$ locations (d) $2 \times 32$ locations

**Q5. One clock cycle of a 2 GHz processor lasts**

(a) 0.5 ns (b) 2 ns
(c) 0.5 µs (d) 2 ms

**Q6. A program that translates and runs one statement at a time is a(n)**

(a) compiler (b) assembler
(c) interpreter (d) linker

**Q7. BIOS/UEFI is an example of**

(a) application software (b) firmware
(c) a utility (d) a device driver

**Q8. Indian bank cheques are read using**

(a) OMR (b) OCR
(c) MICR (d) RFID

**Q9. The QR code was released in 1994 by**

(a) IBM (b) DENSO WAVE
(c) Xerox (d) Intel

**Q10. Pashu Aadhaar uses an ear tag with a**

(a) 6-digit number
(b) 10-digit number
(c) 12-digit bar-coded number
(d) QR code only

**Q11. A solenoid valve controlled by an irrigation controller is a(n)**

(a) input device
(b) actuator (output)
(c) storage device
(d) register

**Q12. A Full HD image of 1920 × 1080 pixels at 24-bit colour needs about**

(a) 2 MB
(b) 6.2 MB
(c) 62 MB
(d) 600 kB

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**Answers**

Q1 — (b) motherboard
Q2 — (c) Program counter
Q3 — (b) fetch, decode, execute, store
Q4 — (b) $2^{32}$ locations — *$2^{32} = 4{,}294{,}967{,}296$.*
Q5 — (a) 0.5 ns — *$1/(2 \times 10^9)$ s.*
Q6 — (c) interpreter
Q7 — (b) firmware
Q8 — (c) MICR
Q9 — (b) DENSO WAVE
Q10 — (c) 12-digit bar-coded number
Q11 — (b) actuator (output)
Q12 — (b) 6.2 MB — *$2{,}073{,}600 \times 3$ bytes.*